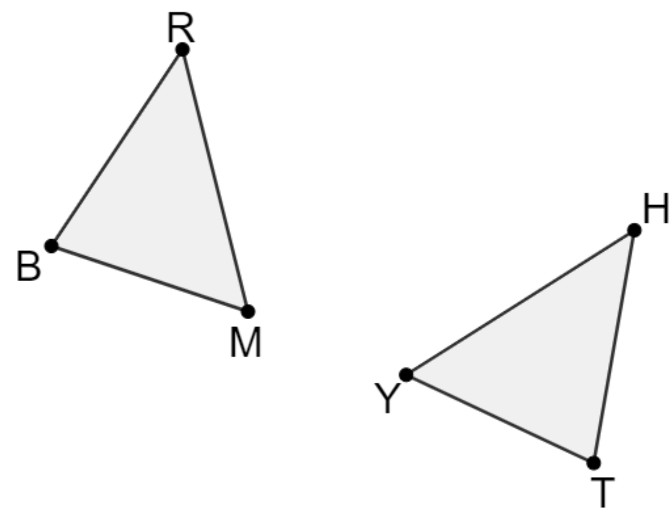
Geometry Honors
Unit 5 Assessment

Name______________________
Date__________________
Period__________

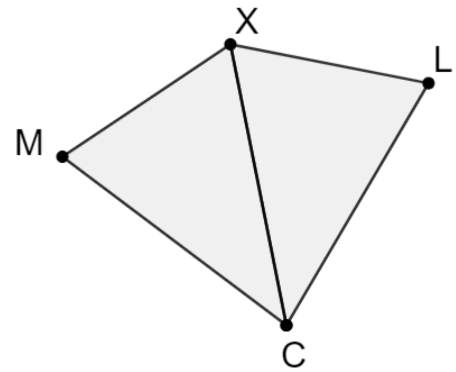
Triangles BRM and YHT are shown. Determine if the given information is sufficient to conclude that the triangles are congruent. If so, name the congruence theorem or postulate that shows congruence.



Given Information	Are They Congruent?	Theorem or Postulate Used
1. $\overline{MR} \cong \overline{YH}$, $\angle HYT \cong \angle BMR$, $\angle BRM \cong \angle THY$	Yes	ASA
2. $\overline{MB} \cong \overline{YT}$, $\overline{MR} \cong \overline{HY}$, $\overline{BR} \cong \overline{TH}$	Yes	SSS
3. $\overline{MB} \cong \overline{YT}$, $\angle YTH \cong \angle MBR$, $\angle BRM \cong \angle THY$	Yes	AAS

4. Quadrilateral $CLXM$ is shown where $\overline{MC} \cong \overline{LC}$ and $\angle MCX \cong \angle XCL$.

Complete the paragraph proof.

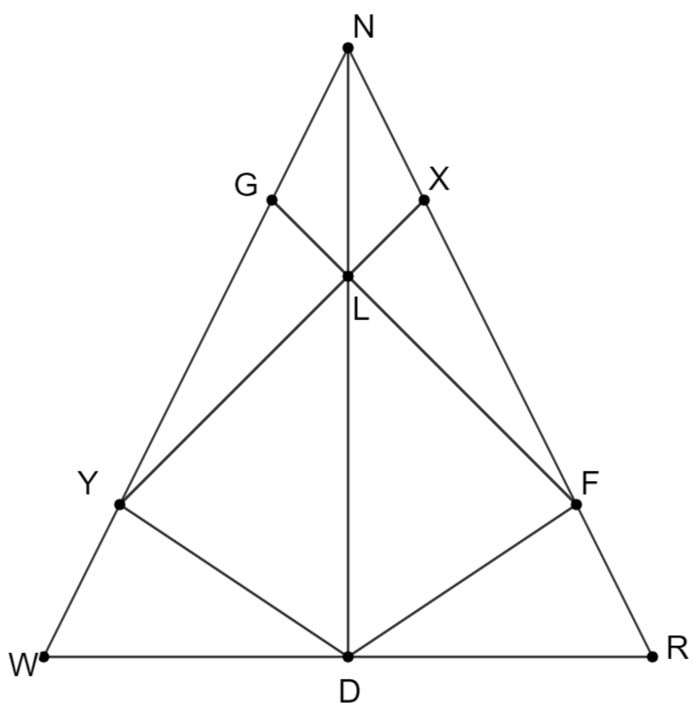


It is given that $\overline{MC} \cong \overline{LC}$ and $\angle MCX \cong \angle XCL$.

By the reflexive property, $\overline{XC} \cong \overline{XC}$. Therefore,

$\Delta MCX \cong \Delta LCX$ by SAS.

5. Triangle NRW is shown.

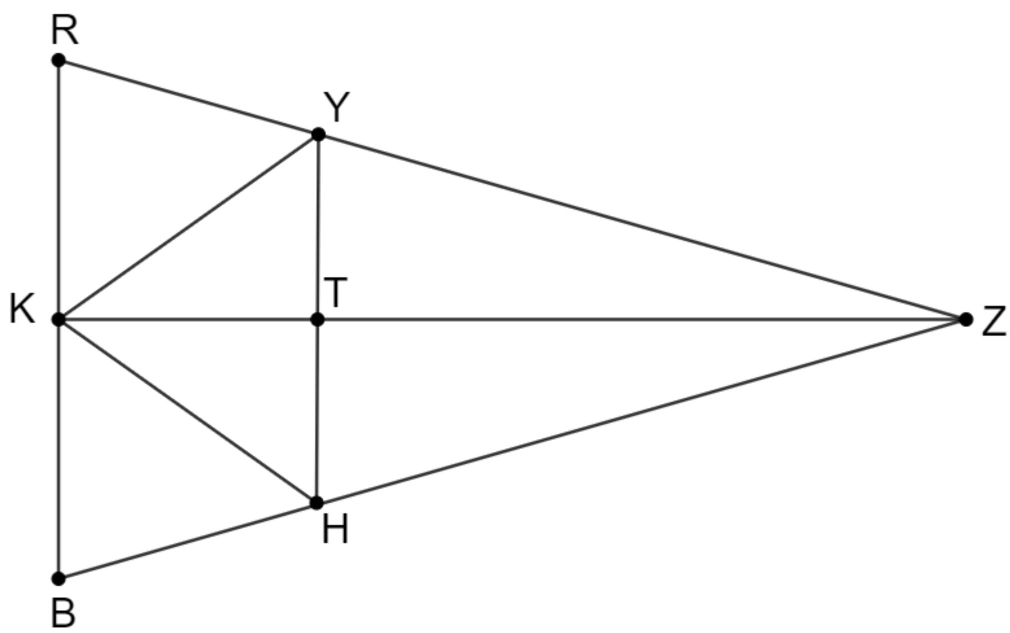


Complete each statement.

a. If $\Delta FLD \cong \Delta YLD$ then $\angle DFL \cong \angle DYL$.

b. If $\overline{ND} \perp \overline{WR}$ and $\overline{ND} \cong \overline{ND}$, the information needed to show that $\Delta NDW \cong \Delta NDR$ using HL is $\overline{NW} \cong \overline{NR}$.

6. Triangle RZB is shown.



Given: $\Delta KYT \cong \Delta KHT$

Prove: $\Delta YKZ \cong \Delta HKZ$

An incomplete proof is shown.

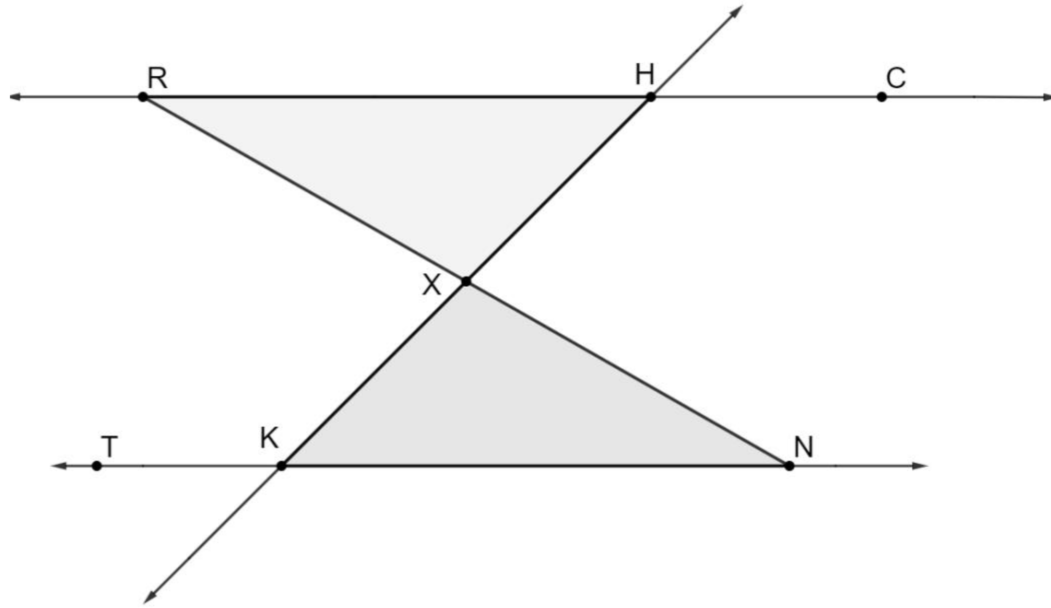
Statement	**Reason**
1. $\Delta KYT \cong \Delta KHT$	1. Given
2. $\overline{KY} \cong \overline{KH}$	2.
3.	3. Corresponding Parts of Congruent Triangles are Congruent (CPCTC)
4.	4. Reflexive Property
5. $\Delta YKZ \cong \Delta HKZ$	5.

What is the reason for step 5?

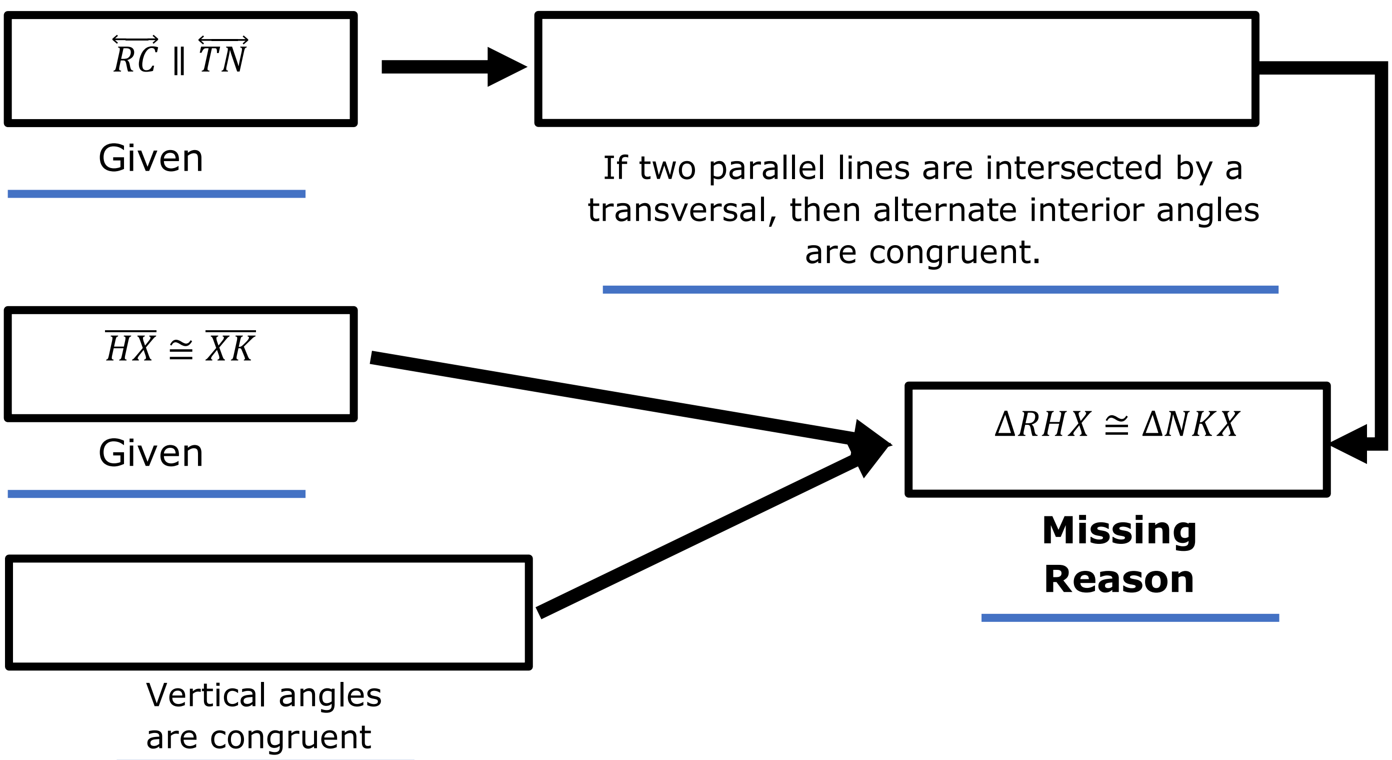
Reason for step 5	SAS

7. Parallel lines $\overleftrightarrow{RC}$ and $\overleftrightarrow{TN}$ with transversal $\overleftrightarrow{KH}$ is shown.

Given: $\overleftrightarrow{RC} \parallel \overleftrightarrow{TN}$,
$\overline{HX} \cong \overline{XK}$
Prove: $\Delta RHX \cong \Delta NKX$



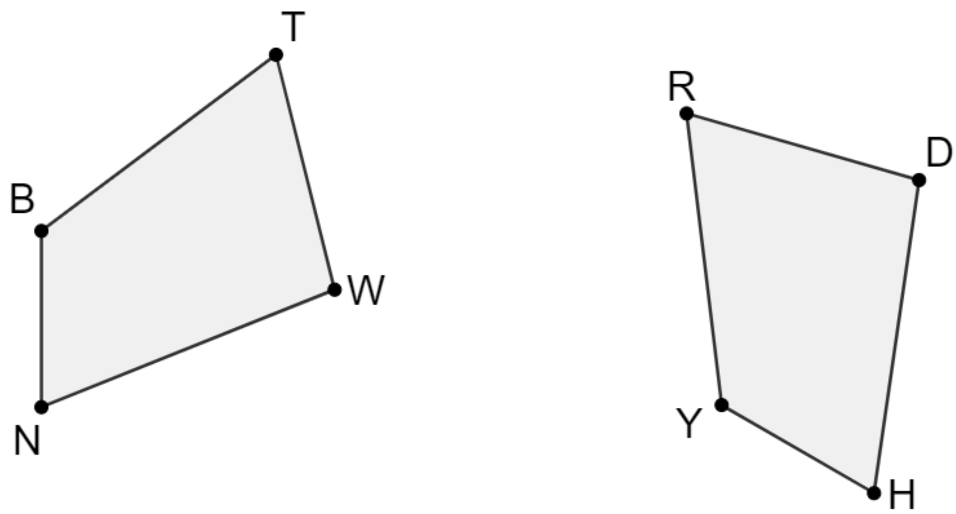
An incomplete flowchart proof is shown.



What is the missing reason in the flowchart proof?

Missing Reason	ASA

8. Quadrilaterals $TWNB$ and $HYRD$ are shown.



It is given that $TWNB \cong RDHY$, $NW = 3x$, $RD = 5x - 19.6$, $RY = 2x + 5.6$, $HD = 5x - 14.4$, $TW = 3x - 5.2$, and $BN = x + 4.8$.

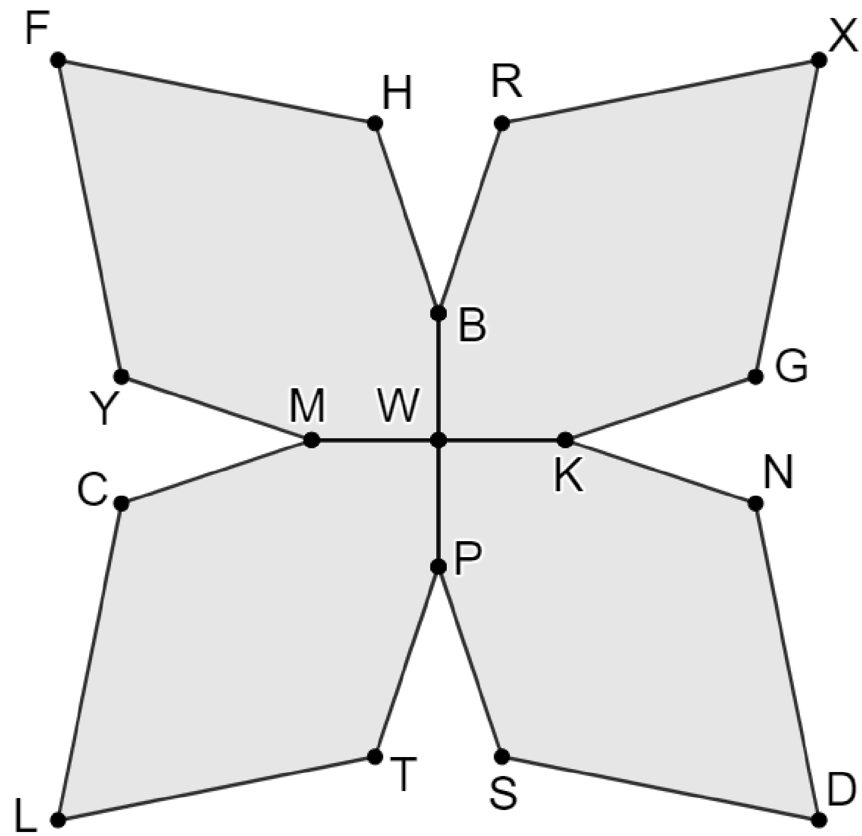
a. Determine the value of x. Show your work.

$x = 7.2$

b. Determine the perimeter of quadrilateral $TWNB$. Show your work.

Perimeter	70

9. A graphic artist created a design for a flower using four congruent hexagons as shown.



Given: $\overline{BP} \perp \overline{MK}$,
$m\angle MYF = m\angle FHB$,
$m\angle WMC = m\angle WPT$,
$m\angle GXR = 66°$,
$m\angle DNK = (3x + 18)°$, and
$m\angle WBR = (5x - 8)°$. The sum of the angles in $WKNDSP$ is $720°$.

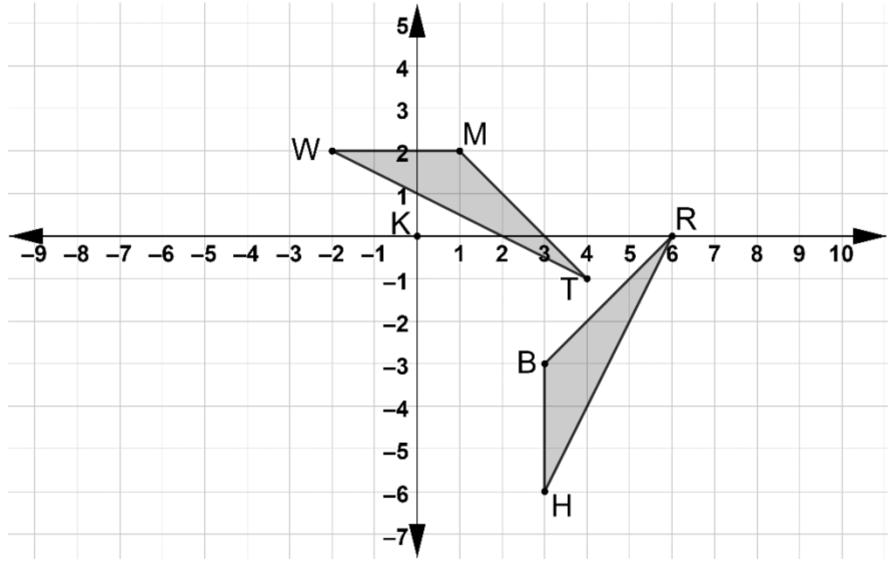
Find each. Show your work

Work

a. $m\angle TLC = 66°$

b. $m\angle KGX = 120°$

10. Triangles TMW and BHR are shown, where $\overline{HB} \cong \overline{WM}$, $\angle WMT \cong \angle HBR$, and $\angle HRB \cong \angle WTM$.



a. Describe the transformations that will map ΔRBH to ΔTMW.

Translate ΔRBH to the left 2 units and down 1 unit, then rotate $90°$ clockwise about point R

b. Complete the statements.

Because the mapping described consists of
- ○ dilations,
- ○ rigid motions,
- ○ both dilations and rigid motions,

the definition of
- ○ similarity
- ○ congruence

can be applied to show that
- ○ the vertices
- ○ corresponding sides
- ○ corresponding angles
- ○ both corresponding sides and corresponding angles

are
- ○ similar.
- ○ congruent.

Therefore, since $\overline{HB} \cong \overline{WM}$, $\angle WMT \cong \angle HBR$, and $\angle HRB \cong \angle WTM$ were given, then
- ○ AAS
- ○ ASA
- ○ SAS
- ○ SSS

is sufficient to show triangles are
- ○ similar.
- ○ congruent.